

Azure Storage Static Website with Front Door

Abstract

This project looks at a simple, low-cost way to host websites using Azure Blob Storage's static website feature, paired with Azure Front Door for global delivery and secure custom domains. For content that doesn't need server-side processing — like department pages, docs, or marketing sites — this setup is a practical alternative to running a full VM or app service.

Static files (HTML, CSS, JS) are served directly from a storage container, so there's no server to manage. Front Door sits in front of it to handle the custom domain, HTTPS certificates, and CDN caching, since Blob Storage on its own can't serve HTTPS on a custom domain. It also adds load balancing and basic security features, making the setup more reliable for real use.

A team of 4 will work on this under tickets T080 and T182, getting hands-on with Azure architecture, CDN setup, DNS, and certificate management — along with the kind of troubleshooting that comes up in real deployments. The goal is a working, globally accessible site on a custom domain with HTTPS, plus a solid understanding of what it takes to build this kind of low-cost cloud hosting setup.

Problem Statement

Small businesses and departments need a professional looking website but have neither the budget nor the capability to set up and manage their own web servers or platform-as-a-service instances. Conventional solutions come with recurrent compute costs, patching and management of servers, as well as concerns of scalability for mostly static websites. Further, delivering these websites securely through a custom domain (HTTPS) and providing good performance for a wide geographic user base would require extra infrastructure including CDNs and certificate management. This solution is required to eliminate server management, minimize cost, and deliver fast, secure and globally available static websites. The proposed solution involves Azure Blob Storage as static website hosting and Azure Front Door.

Objectives

- House a static website with the help of Azure Blob storage without setting up any web servers.
- Maintain Azure Front Door service to utilize custom domain name with HTTPS/TLS.
- Enhance performance and availability with CDN cache at edge nodes.
- Set up routing rules that would be helpful for SPA navigation.
- Record details about implementation, cost considerations, and operational challenges of this architecture.

Use Cases

- House a website for the department in an inexpensive manner, without the requirement of managing and scaling a web server.
- Deploy the website across the globe using HTTPS through the CDN and managed certificates of Front Door.
- Make available portals, landing pages, or marketing microsites, which mostly have static content.
- Serve as a reference architecture for future low-cost and serverless static hosting initiatives.

Technology Stack

- Static Site Hosting with Azure Blob Storage – hosting HTML, CSS, and JavaScript assets.
- Global CDN using Azure Front Door – CDN services and TLS/SSL certificates management.
- DNS services / Domain registration service from Azure – domain configuration and validation.
- Frontend framework (React/Vite, HTML/CSS/JavaScript) – static site or SPA development framework.

Functional Requirements

- FR-01: The system shall host static website files in an Azure Storage account.
- FR-02: The system shall expose the static website through Azure Front Door.
- FR-03: The system shall provide HTTPS access to website users.
- FR-04: The system shall support configuration of a custom domain through Front Door.
- FR-05: The system shall route requests to the Azure Storage static website origin.

Expected Outcome

At the conclusion of the project, it is anticipated that the team would have created a static site running on the Azure Blob storage service which can be accessed via a globally-accessible custom domain name via the Azure Front Door Service using HTTPS. This should show proper single-page app routing, CDN caching, and an easy way to recreate the same.